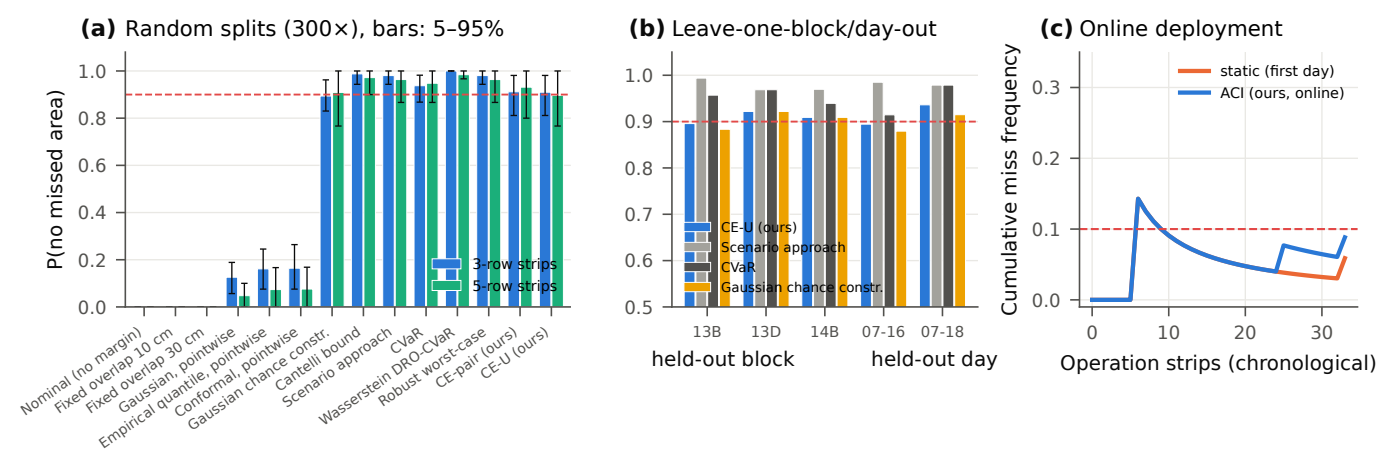
(a) Random splits (300×), bars: 5–95%
P(no missed area)
1.0
0.8
0.6
0.4
0.2
0.0
3-row strips
5-row strips
Nominal (no margin)
Fixed overlap 10 cm
Fixed overlap 30 cm
Gaussian, pointwise
Empirical quantile, pointwise
Conformal, pointwise
Gaussian chance constr.
Cantelli bound
Scenario approach
CVaR
Wasserstein DRO-CVaR
Robust worst-case
CE-pair (ours)
CE-U (ours)
(b) Leave-one-block/day-out
1.0
0.9
0.8
0.7
0.6
0.5
CE-U (ours)
Scenario approach
CVaR
Gaussian chance constr.
13B
13D
14B
07-16
07-18
held-out block
held-out day
(c) Online deployment
Cumulative miss frequency
0.3
0.2
0.1
0.0
static (first day)
ACI (ours, online)
0
10
20
30
Operation strips (chronological)